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Assignment-8

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1. Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

❖ Steps to Assign a Static IPv6 Address (Windows)

Step 1: Open Command Prompt as Administrator.

Step 2: Find your network adapter name.

ipconfig

Step 3: Assign a static IPv6 address (replace Wi-Fi with your adapter name if needed).

netsh interface ipv6 set address interface="Wi-Fi" address=2001:db8:1::10

Step 4: Verify the address.

ipconfig

Result: The IPv6 address should appear under your Wi-Fi adapter.

Note: 2001:db8:1::10 is for practice only, not for real internet connectivity.

❖ Steps to Assign a Static IPv6 Address in Linux

Step 1: Open the Terminal.

Step 2: Check your network adapter.

ip -6 addr

Step 3: Assign a static IPv6 address (replace eth0 with your adapter name).

sudo ip -6 addr add 2001:db8:1::10/64 dev eth0

Step 4: Verify the address.

ip -6 addr show dev eth0

Result: Your IPv6 address should appear in the output.

Note: This is a temporary configuration and the example address is for practice only. It will not provide real internet connectivity.

2. Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.Hint: Use the ping6 or ping -6 command for IPv6 testing.

❖ Steps to Configure Unique Local IPv6 in Linux

Step 1: Check your network interface

ip link

Step 2: Assign a unique local IPv6 address Replace eth0 with your interface name.

sudo ip -6 addr add fd12:3456::10/64 dev eth0

Step 3: Configure another LAN device

Assign the second device:

sudo ip -6 addr add fd12:3456::20/64 dev eth0

Step 4: Test connectivity from the first device

ping -6 fd12:3456::20

Result: If you receive replies, both devices can communicate using IPv6.

Note: Use the actual interface name on each device. These addresses are examples; both devices must be on the same LAN, and firewalls must allow IPv6 ping.

3. Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.Constraint: Document the steps and the observed address behavior in a text file.

Report :

Temporary IPv6 Address Configuration and Verification

Objective Set a temporary IPv6 address and check whether it remains after a reboot or network disconnect/reconnect.

Linux Steps 1. Open Terminal. 2. Find the network interface: ip link 3. Check current IPv6 addresses: ip -6 addr 4. Add a temporary IPv6 address (replace eth0 with your interface name): sudo ip -6 addr add fd12:3456::10/64 dev eth0 5. Verify the address: ip -6 addr show dev eth0 6. Remove and re-add the address, or disconnect/reconnect the network, then check again: ip -6 addr show dev eth0 7. Reboot the system and check again: ip -6 addr show dev eth0

Expected Behavior An address added with the ip addr add command is temporary in the sense that it is not saved in the persistent network configuration. It normally disappears after a reboot. A network disconnect/reconnect may also remove it, depending on the network manager and interface behavior.

Observed Results
- Interface name: enp0s3
- Address shown immediately after adding it: fd12:3456::10/64
- After disconnect/reconnect: Address may disappear or remain, depending on the system.
- After reboot: Address normally disappears because it was not permanently configured.

Conclusion Record whether the address remained or disappeared at each check. The actual result can vary depending on the Linux distribution and network manager.

Note The example address fd12:3456::10/64 is a documentation/lab example for a unique local IPv6 address. Use an address that does not conflict with another device on your LAN. These commands do not configure a default route or guarantee internet connectivity.

4. Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

Here is a simple Python script for a Zomato-style restaurant search.

```
restaurants = {
    "Pizza House": "fd12:3456::1",
    "Burger King": "fd12:3456::2",
    "Dominos Pizza": "fd12:3456::3",
```

```
    "Food Court": "fd12:3456::4"
}
name = input("Enter restaurant name: ").lower()
for restaurant, ip in restaurants.items():
    if name in restaurant.lower():
        print(restaurant, ":", ip)
```

Example Output

```
Enter restaurant name: pizza
Pizza House : fd12:3456::1
Dominos Pizza : fd12:3456::3
```

How it works:

1. Stores restaurant names and IPv6 addresses.
2. Takes a partial restaurant name from the user.
3. Searches for matching names.
4. Displays the matching restaurant's IPv6 address.

5. Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

IPv6 Subnetting Scenario and Manual Configuration

1. AI-Generated IPv6 Subnetting Scenario
   A small food delivery company needs three separate IPv6 subnets: - Delivery Partners: 2001:db8:100:1::/64 - Restaurants: 2001:db8:100:2::/64 - Admin Office: 2001:db8:100:3::/64

   Example host addresses: - Delivery Partners system: 2001:db8:100:1::10/64 - Restaurant system: 2001:db8:100:2::10/64 - Admin system: 2001:db8:100:3::10/64

   Note: The 2001:db8::/32 prefix is reserved for documentation and examples. These addresses are for a lab demonstration and will not provide real Internet connectivity.

2. Linux Configuration Steps

   Step 1: Open Terminal.
   Step 2: Find your network interface: ip link
   Step 3: Add the sample IPv6 address to your interface (replace enp0s3 with your actual interface name): sudo ip -6 addr add 2001:db8:100:1::10/64 dev enp0s3

   Step 4: Verify the address: ip -6 addr show dev enp0s3
   Step 5: Record the result shown by your system: Interface name: enp0s3
   IPv6 address visible: 2001:db8:100:1::10/64

3. Windows Alternative

Step 1: Open Command Prompt as Administrator.

Step 2: Find the adapter name: netsh interface ipv6 show interfaces

Step 3: Add the sample address (replace “Wi-Fi” with your adapter name): netsh interface ipv6 add address interface=“Wi-Fi” address=2001:db8:100:1::10/64

Step 4: Verify: ipconfig

4. Important Notes

-Choose only one operating system’s steps.

- Replace the interface/adapter name with the actual name on your system.

- This is a manual, temporary configuration and may disappear after reboot or network changes.

- Do not use the documentation prefix on a production network.

- If the command fails or the address does not appear, record the actual error and check the interface name and permissions.